

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 18: Texture Analysis

WORKSHEET SOLUTIONS

1. 1. Match the GLCM (Gray-Level Co-occurrence Matrix) feature to its meaning:

GLCM Feature	Matched Meaning / Mathematical Interpretation
Contrast	Measures local intensity variations between a pixel and its neighbor over the image (intensity contrast).
Entropy	Measures the randomness, disorder, or complexity of the spatial gray-level distribution.
Energy (Uniformity / ASM)	Measures textual uniformity or orderliness (sum of squared elements in the GLCM).
Homogeneity (Inverse Difference Moment)	Measures the closeness of the distribution of elements in GLCM to the diagonal (local smooth similarity).

2. 2. Which texture type has high entropy? Why?

Rough / Complex / Random Textures (such as sand, noisy surfaces, woven fabric, or highly detailed natural patterns) have high entropy.

Reason: Entropy quantifies the amount of spatial randomness or unpredictability in pixel intensity distributions. In a complex or rough texture, gray-level co-occurrences are distributed almost uniformly across many entries in the GLCM rather than being concentrated in a few specific pairs, resulting in higher information disorder ($-\sum P_{i,j} \log_2 P_{i,j}$).